GENERALGUIDELINES

Do's:-

- Students should be on time for every lecture.
- Students are advised to show due respect to all faculty members.
- Students should keep the Classrooms, Laboratories and Workshops clean and tidy.
- Students must maintain absoluted is cipline and decorum, while on campus.
- **Students should come prepared with algorithm / flowchart / program / procedure for all the experiments before attending the laboratory session.**
- Students are advised to clarify their doubts in the respective courses with the faculty.
- Students have to inform their parents that they should follow up the progress of their wards by being in touch with the institution authorities at regular intervals.
- **Students are advised to be present for the mentor meetings conducted by their respective Faculty Advisors, failing which appropriate disciplinary action will be taken.**

Don'ts:-

- Students are not permitted to attend the class without the identity card, once issued.
- Ragging is strictly prohibited because it is punishable under Karnataka Education Act. Any student involved in ragging, will be severely punished –which includes handing over the case to Police, rustication from the college etc.
- Writing on desks and walls is strictly prohibited, failing which the students will be fined heavily. If the identity of the individual is not established the entire class / students in the block will be fined.
- Students must not use their cell phones during class hours. If any student is found using their cell phone during class hours it will be confiscated.
- Students are not supposed to alter the configuration of the system/any software on the systems.

IV SEMESTER (2024-28 BATCH)

Sl. No.	Course Code	Course Title	Hours per week				Credits	AI Tools/ Tools/Languages	Course Type
			L	T	P	S	C		
1	UE24CS251B	Microprocessor and Computer Architecture	4	0	2	5	5	ARM Simulator, Arduino Microcontroller kit, Para Cache simulator. Compute intensive Application are run on GPU systems	CC-Lab Integrated
2	UE24CS252B	Computer Networks	4	0	2	5	5	Wireshark, Python. Advanced-Computer Networking-Tutor Computer Network Professor (yeschat.ai)AI	CC-Lab Integrated
3	UE24CS241B	Design and Analysis of Algorithms	4	0	0	4	4	C-Programming language Visu Algo (Interactive Visualizations), Algorithm Visualizer (AI Explanations)	CC-Independent
4	UE24CS242B	Operating Systems	4	0	0	4	4	C, Linux/ Unix OS for system call implementation	CC-Independent
5	UE24MA241B	Linear Algebra and its Applications	4	0	0	4	4	Python IDE, MathGPT, Deepseek	CC-Independent
6	UE24EC221B	CIEL2	2	0	0	2	2		CC-Independent
7	UE25MA221B	Bridge Course Mathematics–II (Applicable to Lateral Entry Students)	2	0	0	2	0		FC-Independent
Total			**24/26**	**0**	**4**	**24/26**	**24**		

UE24CS251B -Microprocessor and Computer Architecture

Class	Units	Reference Book & Chapter #	Topics	% of syllabus	Cumulative
1	Unit 1	T3: Chapter 1, Section 1.1	Introduction, ISA Classification - RISC and CISC	28%	28%
2		R2: Chapter 1, Section 1.5, Chapter 2, Section 2.1-2.3	Processor design trade-offs, ARM Processor Architecture		
3		T3: Chapter 2, Section 2.1, 2.2	Register set, Introduction to ARM ISA		
4		T3: Chapter 3, Section 3.1-3.5 (Exclude 3.5.1, 3.5.2),	Data Processing Instructions: Addition and Subtraction with programming examples,		
5 to 6		Lab 1: ARM7TDMI : ISA Program Sequence 1			
7		T3: Chapter 3, Section 3.1-3.5 (Exclude 3.5.1, 3.5.2),	Data Processing Instruction Variants, Branch Instructions		
8			Addressing Modes		
9			Data Transfer Instructions		
10			Load and Store with programming examples		
11 to 12		Lab 2: ARM7TDMI : ISA Program Sequence 2			
13		T3: Chapter 4: Section 4.3	Conditional Execution		
14			Multiplication Instructions		
15			Load Multiple and Store Multiple Instructions		
16			Interrupts and Programming Examples		
17-18		Lab 3: ARM7TDMI : ISA Program Sequence 3			
19		R2: Chapter 2,Chapter 3, Chapter 5 : 5.1 - 5.15	Interrupts, Context Switching an Overview.		
20			Functions, Parameter Passing Techniques		
21			Instruction Encoding 1-2: Data Processing and Data Transfer Instructions		
22			Instruction Encoding 3: Branch and Other Instructions		
23-24		Lab 4:ARM7TDMI : ISA Program Sequence 4			

UE24CS251B -Microprocessor and Computer Architecture

25	Unit 2	T2 : Chapter 4 Section: 4.5, 4.7, 4.8	Introduction to Pipelining (3 & 5 Stage)	22%	50%
26			Introduction to Pipeline Hazards		
27			Structural Hazards		
28			Data Hazards		
29-30		Lab 5: ARM7TDMI : ISA Program Sequence 5			
31		T2 : Chapter 4 Section: 4.5, 4.7, 4.8	Data Hazards : Examples and Problems		
32			Control Hazard		
33			Performance Analysis-1		
34			Performance Analysis-2, Performance Analysis Problems		
35-36		Lab 6: Banana Problem (Unit 1 & 2)			
37		R1 : Appendix C-1, C-2, C-3	Introduction to Branch Prediction		
38			1 bit : Branch Prediction		
39			2 bit: Branch Prediction		
40			Branch Prediction Problems		
41-42		Lab 7: ARM7TDMI : ISA Plugin Programs			
ISA - 1					
43	Unit 3	R1: Appendix B - 1, B - 2, B -3,	Introduction to Memory Hierarchy	25%	75%
44			Basics of Caches, Fully Associative		
45			Direct Mapped, Set Associativity		
46			Cache Performance		
47-48		Lab 8: ARM7TDMI : ISA Plugin Programs			
49		R1: Appendix B - 1, B - 2, B -3,	Basic Cache Optimization		
50			Reduction in Miss Rate		
51			1st Optimization and examples		

UE24CS251B -Microprocessor and Computer Architecture

52			2nd Optimization and examples		
53-54		Lab 9: Mini Project Selection and Finalisation, Orange Problem			
55		R1: Appendix B - 1, B - 2, B -3,	3rd Optimization and examples		
56			Write Policies		
57			Write Policies		
58			4th Optimization with examples		
59-60		Lab 10: Orange Problem Evaluation			
61		R1: Section: 2.1, 2.2 upto page 87	5th Optimization	25%	100%
62			6th Optimization		
63	Unit 4	T4: Chapter 1, section 1.1 - 1.3, 1.5,	Introduction to Parallel Computing		
64			Parallel Computer Memory Architecture		
65-66		Lab 11: Mini Project Review 1			
67		T5: Chapter 1: Section 1.1 - 1.2, 1.4 - 1.5	Flynn's Taxonomy		
68			Parallel Computing Concepts and Terminology		
69			Design Issues & Constraints		
70			Performance in Parallel Computing : Amdahl's Law		
71-72		Lab 12: Mini Project Final Review			
73		T5: Chapter 1: Section 1.1 - 1.2, 1.4 - 1.5	Performance in Parallel Computing : Amdahl's Law		
74			Gustafson's Law		
75			Multicore Processors and OpenMP Programming		
76			Multicore Processors and OpenMP Programming		
77-78		Lab 13: Banana Problem (Unit 3 & 4)			
79		Hand-outs	Realization of ML operations on Modern Architecture		
80			Introduction to Heterogenous Parallel Computing		

UE24CS251B -Microprocessor and Computer Architecture

81			Introduction to Heterogenous Parallel Computing		
82			Introduction to GPU Computing		
83-84		Lab 14: Mini Project Final Review (Contd.)			
ISA - 2					

#	Category	# of Sessions	%
1	Lectures Hours	56	66.67%
2	Experiential Learning		
2a	Banana Problem	4	16.67%
2b	Orange Problem	4	
2c	Jackfruit Problem	6	
3	Lab sessions	14	16.67%
	Total	84	100%

Book	Code	Title and Author	Edition	Publisher	Year
Text Book	T1	“Computer Organization and Design”, Patterson,	Fifth Edition	Elsevier	2014
Text Book	T2	“Computer Organization and Design”, Patterson,	Fourth Edition	Elsevier	2012
Text Book	T3	“ARM System Developer's Guide Designing and Optimization System Software”, Andrew N Sloss et al.,		Elsevier	2004
Text Book	T4	“Programming Massively Parallel Processors A	Third Edition	Elsevier	2017
Text Book	T5	“Multicore and GPU Programming An	Second Edition	Elsevier	2015
Reference Book	R1	“Computer Architecture: A Quantitative Approach”,	Fifth Edition	Elsevier	2011
Reference Book	R2	“ARM System-on-Chip Architecture”, Steve Furber	Second Edition	Pearson Educatio	2015

UE24CS241B-Design and Analysis of Algorithms

Class #	Chapter Title /Reference Literature	Topics to be Covered	% of Portion covered	
			Reference Chapter	Cumulative%
1	**Unit #1 T1 : 1.1-1.3, 2.1-2.4, 3.1,3.2,3.4**	The motivation for the course. Evaluation policy of the course. Introduction to Algorithms.	25	25
2		Fundamentals of Algorithmic problem-solving. Important problem types – sorting, searching.		
3		Important problem types – string processing, graph problems, Combinatorial, Geometric, numerical problems		
4		Analysis Framework, Orders of Growth		
5		Asymptotic Notations, Basic Efficiency Classes		
6		Using Limits for comparing order of growth		
7		Mathematical Analysis of Non-recursive Algorithms		
8		Mathematical Analysisof Recursive Algorithms		
9		Solving Recurrences of Recursive Algorithms		
10		Brute-Force approach, selection sort		
11		Bubble sort, Sequential Search, Brute-Force String Matching,		
12		Exhaustive Search–Travelling Salesman Problem, Knapsack Problem, Assignment Problem		
13		**Assignment 1 – Unit 1**		
14		**Revision**		
15	**Unit #2 T1: 4.1 – 4.5, 5.1-5.4**	Decrease-and-Conquer approach-Insertion Sort	25	50
16		Topological Sorting		
17		Algorithms for Generating Combinatorial, Objects: Generating Permutations , Johnson Trotter Algorithm		
18		Generating Permutations in Lexicographic order		
19		Generating Subsets		
20		Decrease-by-a-Constant-Factor Algorithms: Fake coin Problem Russian Peasant Method for Multiplication, Josephus Problem		
21		Decrease by Variable size algorithms-Computing median and selection Problem.		
22		Divide and Conquer Approach, General Divide and Conquer Recurrence,		

		Master Theorem,		
23		Merge Sort, Merge Sort Analysis(Best case, Worst case		
24		Quick Sort, Quick Sort Analysis(Best case, Worst case)		
25		Binary Search, Binary Tree Traversals, Multiplication of Large Integers		
26		Strassen's Matrix Multiplication		
27		**Assignment -2:Unit 2**		
28		**Revision**		
29	**Unit #3** **T1: 6.1,6.3,6.4** **7.1,7.2,7.4** **9.1-9.4** **R1-18.1,18.2**	Transform-and-Conquer Approach Pre-sorting, Heap Sort	25	75
30		Red-Black Trees		
31		2-3 Trees,		
33		B Trees		
34		Space and Time Tradeoffs-Sorting by Counting, Distribution Counting Sort		
35		Input Enhancement in String Matching– Horspool's algorithm,		
36		Boyer-Moore Algorithm		
37		Boyer-Moore Algorithm cntd		
38		**Assignment 3:Transform and Conquer, Space and Time Tradeoffs (Hands on)**		
39		Greedy Technique, Prim's Algorithm		
40		Kruskal's Algorithm, Kruskal's Algorithm (union and find algorithm)		
41		Dijkstra's Algorithm		
42		Huffman trees		
43		Revision		
44	**Unit #4** **T1 : 81.,8.2,8.4** **11.1-11.3** **12.1,12.2**	**Uploading of Jackfruit Problem- Group Assignment.** Dynamic Programming: Computing a Binomial Coefficient		
45		The Knapsack Problem , Memory Functions for solving Knapsack Problem		
46		Warshall's Algorithm to find Transitive Closure		
47		Floyd's Algorithm for All Pair Shortest path problem		
48		**Assignment -4 –Dynamic Programming Greedy Method (Hands on)**		
49		Limitations of Algorithm Power: Lower-Bound Arguments, Decision Trees		
50		P, NP, and NP-Complete, NP-Hard Problems		
51		Coping with the Limitations of Algorithm Power, Backtracking: N Queen, Subset sum problem		
52		Branch-and-Bound: TSP		
53		Branch and Bound: knapsack		
54		Branch and Bound: Job Assignment problem		
55		**Revision**		
56		**Revision**		

UE24CS241B-Design and Analysis of Algorithms

Programming Language C (GCC compiler)

Text Book(s):
1: "Introduction to the Design and Analysis of Algorithms", Anany Levitin, Pearson Education, Delhi (Indian Version), 3rd Edition, 2012.

Reference Book(s):
1: "Introduction to Algorithms", Thomas H Cormen, Charles E Leiserson, Ronald L Rivest and Clifford Stein, Prentice-
HallIndia,3rdEdition,2009.
2:"Fundamentals of Computer Algorithms", Horowitz, Sahni, Rajasekaran,Universities Press,2nd Edition,2007. 3: "Algorithm Design", Jon Kleinberg, Eva Tardos, Pearson Education, 1st Edition, 2006